

Multiple Choice (10 marks)

1. How many integers between 1 (included) and 100 (included) are divisible by either 2, 3, or 5?
 - (a) 82
 - (b) 60
 - (c) 80
 - (d) 68
 - (e) 74
2. For how many positive integers k does the ordinary decimal representation of the integer $k!$ end in exactly 99 zeros?
 - (a) Twenty
 - (b) Three
 - (c) None
 - (d) Four
 - (e) Five
3. For $a \geq 0$, we define $S_a = \{x \mid \text{dist}(x, S) \leq a\}$, where $\text{dist}(x, S) = \inf_{y \in S} \|x - y\|$. Suppose S is convex. Is S_a convex? Return 1 for yes and 0 for no.
 - (a) Undefined
 - (b) 0.0
 - (c) None of the above
 - (d) 1.0
4. At West Elementary School, there are 20 more girls than boys. If there are 180 girls, how can you find the number of boys?
 - (a) divide 20 by 180
 - (b) add 180 to 20
 - (c) multiply 180 by 20
 - (d) multiply 180 by 2
 - (e) divide 180 by 20
5. We know that $y' = (x+y)/2$, we also know that $y(x=0) = 2, y(x=0.5) = 2.636, y(x=1) = 3.595, y(x=1.5) = 4.9868$, what is the value of $y(2)$ using Adams bashforth predictor method.

- (a) 5.4321
- (b) 7.1234
- (c) 6.2456
- (d) 7.7654
- (e) 5.6789

True / False (10 marks)

Answer True or False

6. True or False: The value of the expression $-16 + 13 + (-33)$ simplifies to -4 .
7. True or False: The correct answer to 'Given that $n > 1$, what is the smallest positive integer n whose positive divisors have a product of n^6 ?' is '10'.
8. True or False: There is only one group of order n (up to isomorphism) for the integers $n = 3, 5, 7$, and 11 .
9. True or False: The probability that the sample mean falls between 74.2 and 78.4 for a normal distribution $N(77, 25)$ is 0.6398 .
10. True or False: The composition $f(g(h(x)))$ simplifies to $-15x^2 + 24$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the mathematical approach to calculating the total time Susan would take to read 18 pages at her consistent reading rate of 1 page every 3 minutes. Detail the steps involved in your analysis.
12. Calculate the new waist size Owen must wear if his waist circumference increases by 15% from his original 32-inch size. Discuss the mathematical process behind your calculations.

End of Paper